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RL Post-Training Builds Compositional Reasoning Strategies

UCL Gatsby Researchers Demonstrate That RL with Only Binary Reward Can Compose Pre-trained Primitive Skills Into New Higher-Level Strategies That Rejection Fine-Tuning Cannot Reach

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A central question in post-training research is whether reinforcement learning merely amplifies skills already latent in a pretrained base model, or whether it can build genuinely new capabilities by composing those primitives in ways the model has never seen. Researchers at UCL’s Gatsby Computational Neuroscience Unit now answer this question decisively using a controlled symbol-rewrite grammar environment: RL with binary final-answer reward not only outperforms rejection fine-tuning on composite tasks, it produces a new compositional mechanism that emerges in two identifiable phases.

AT A GLANCE

Problem: It is unknown whether RL post-training amplifies existing skills or composes them into genuinely new strategies unavailable to the base model.

Why it matters: If RL can only amplify, it is bounded by pretraining; if it can compose, it can transcend pretraining data in principled ways.

Method: Controlled rewrite-grammar environment where primitives and compositions are known; Transformer pretrained on primitives, RL post-trained on composite tasks with binary reward.

Result: RL reaches 78% on composite problems vs. 40% for RFT plateau; compositional strategies emerge in two phases.

Evidence: Phase transition at step ~600; sequential compositions emerge first, then parallel compositions.

The Big Picture

The dominant post-training recipe for reasoning models involves either rejection fine-tuning (RFT) — filtering model rollouts for correct answers and training on them — or reinforcement learning (RL) with verifiable reward.

These two approaches are widely believed to have different capability profiles: RFT is seen as a data-efficient curriculum tool that stays within the distribution of the base model, while RL is believed to discover new strategies through trial and error. But the evidence for this distinction has been largely anecdotal.

The Gatsby team designs a setting where ground truth is available for exactly which strategies the model uses at each step, making it possible to verify mechanistically whether RL produces new compositional behavior or merely amplifies existing primitives.

Why Existing Approaches Fall Short

Both existing post-training approaches have limitations that are difficult to diagnose on open-ended language tasks. RFT cannot improve beyond the base model’s rollout distribution: if the base model rarely produces composite solutions, RFT has no signal to amplify. RL can in principle explore beyond the base distribution, but it is unclear whether the gradient signal from binary reward is sufficient to wire together primitive capabilities into structured compositions, or whether it simply boosts the probability of already-composite rollouts that happen to appear at low frequency.

In language settings, these mechanisms are entangled with content and fluency; the rewrite-grammar setting surgically isolates the compositional question from all other confounds.

Core Mechanism

EcoSpec’s RL training is shown to operate through a two-phase compositional mechanism. In **Phase 1**, RL sharpens the model’s execution of individual primitive rules: the probability of applying the correct primitive to the correct symbol increases, reducing errors in primitive steps. In **Phase 2**, the model begins chaining these primitive applications — first sequentially (applying rule r_i then r_j in a coordinated trace), then in parallel (applying r_i and r_j to independent subterms simultaneously within a single generation). The transition between phases is sharp and reproducible across seeds.

Technical Formulation

The rewrite grammar defines a set of primitive rules $\mathcal{R} = \{r_1, \dots, r_K\}$. Each primitive r_i rewrites a specific input symbol pattern $s_{\text{in}}^{(i)}$ to $s_{\text{out}}^{(i)}$. A composite problem of arity 2 requires applying rules r_i and r_j either sequentially or in parallel:

Sequential: $r_j(r_i(x)) = y$ Parallel: $(r_i \parallel r_j)(x_1, x_2) = (y_1, y_2)$

The Transformer is pretrained only on single-rule traces (input, output pairs with r_i applied once). At post-

training, it receives composite problem inputs and binary reward for producing the correct final output.

The policy gradient update under REINFORCE takes the form:

$$\nabla_{\theta} \mathcal{J}(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[R(\tau) \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]$$

where $R(\tau) \in \{0, 1\}$ is the binary correctness reward for the full trace τ and the sum is over all generation steps.

Learning Procedure

1. **Pretraining:** The Transformer is trained on a corpus of single-rule rewrite traces covering all K primitives uniformly. After pretraining, the model reliably applies any single primitive but has near-zero accuracy on composite tasks.
2. **RL post-training:** REINFORCE with binary reward is applied to composite tasks. No intermediate-step reward, no step annotations. Training runs for 2000 gradient steps across all conditions.
3. **RFT baseline:** Rollouts from the pretrained model on composite tasks are filtered for correctness; the model is fine-tuned on the filtered set. RFT training uses the same FLOPs as RL where possible.

Experimental Findings

Method	Accuracy	Plateau
Base model (pass@1)	4.3%	—
Base model (pass@100)	19.1%	—
RFT	40.2%	Step 400
RL (this work)	78.3%	Step 2000

The phase transition is visible in trace analysis: the rate of sequential compositions rises from near-zero to $\sim 30\%$ by step 600, then stabilizes while parallel compositions rise from near-zero to $\sim 45\%$ by step 2000.

The base model’s pass@100 rate (19.1%) establishes a ceiling for RFT: RFT cannot learn from solutions the model never produces under any sampling strategy, yet RFT still substantially surpasses this baseline — suggesting that rare composite rollouts provide sufficient training signal when filtered correctly.

Ablations and Interpretation

Sequential before parallel: Parallel compositions never emerge before sequential ones across any random seed or grammar instantiation. The authors interpret this as a task-induced curriculum: sequential composition is a prerequisite for parallel composition because it requires the model to coordinate rule application across a linear chain before coordinating across independent subterms.

Binary vs. step-level reward: Providing step-level intermediate reward speeds up phase 1 (primitive sharpening) but does not accelerate phase 2 (compositional emergence), suggesting that compositional emergence is driven by global reward signal optimizing the full trace rather than local per-step feedback.

RFT saturation: RFT saturates because the base model’s composite-solution rollout rate is low, providing a thin training signal. Once that signal is exhausted, gradient updates from the sparse correct rollouts no longer produce the global policy restructuring that RL achieves through exploration.

Implications: These results support a view of RL as a *composer* rather than merely an *amplifier*: given primitives in pretraining and composition problems at post-training, RL can discover how to chain those primitives into strategies that the base model has never executed.

Reference

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